

t-SNE Project Warmup Exercise

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1 Setup

This document is dedicated to the reconstruction of [1] with low-dimension kernel g replaced by a more general kernel. This would serve as a great way to understand the material and help with the completion of the original project. For the sake of simplicity, we will use the exact same notation as in [1] and skip all the definitions. Recall in [1], g is defined as

$$g(y, y') = \frac{1}{1 + \|y - y'\|^2} \quad (1.1)$$

With all these being said, we will first start by giving our general low dimension kernel g some general assumptions, and modify it as we move on. We first require that $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing.
2. k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' > \infty \quad (1.2)$$

We will kick off this warmup by listing all the important lemmas and results in order, and prove our own versions when the need arises. The final goal is to prove a same set of results as in [1] with the y kernel replaced. Recall that

$$\begin{aligned} F_{\rho, \mu}(x, \sigma) = & - \int \frac{\|x - x'\|^2}{2\sigma^2} \frac{\exp(-\|x - x'\|^2 / 2\sigma^2)}{\int \exp(-\|x - x'\|^2 / 2\sigma^2) d\mu(x')} d\mu(x') \\ & - \log \left(\int \exp(-\|x - x'\|^2 / 2\sigma^2) \right) + \log \rho. \end{aligned} \quad (1.3)$$

Proposition 1. *If μ has a C^1 density f such that both f and $\|\nabla f\|$ are bounded, then the equation $F_{\rho, \mu}(x, \sigma) = 0$ has a unique solution $\sigma_{\rho, \mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$. Moreover, $x \mapsto \sigma_{\rho, \mu}^*(x)$ is smooth.*

Proof. This proposition is about the kernel of x . The proof stays unchanged from [1]. □

Lemma 1. *Let $\delta > 0$ be constant, $K_n = [\delta, \sqrt{n}]$ and $h : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$. Suppose the following hold:*

1. μ is sub-Gaussian;
2. $h(x, x', \sigma)$ and $\nabla_x h(x, x', \sigma)$ are bounded for all x, x' and all $\delta \in K_n$;
3. $h(x, x', \sigma) \rightarrow 0$ as $\|x - x'\| / \sigma \rightarrow \infty$.

Let $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Let $\mu_n = (1/n) \sum_i \delta_{X_i}$. Then for any $\epsilon > 0$ with probability at least $1 - e^{-c\epsilon^2 n}$ we have

$$\sup_{\sigma \in K_{n,x}} \left| \int h(x, x', \sigma) d\mu_n(x) - \int h(x, x', \sigma) d\mu(x') \right| \leq \epsilon + O\left(\sqrt{\frac{\log n}{n}}\right). \quad (1.4)$$

Recall that X or the corresponding probability measure μ is called sub-Gaussian if $\mathbb{P}(X > t) \leq 2 \exp(-Ct^2)$ for some positive constant C .

References

- [1] Antonio Auffinger and Daniel Fletcher. Equilibrium distributions for t-distributed stochastic neighbour embedding, 2023, 2304.03727. URL <https://arxiv.org/abs/2304.03727>.
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